

PalmSource®

Conference and Expo



Lab – PalmDebugger

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PalmDebugger is

- A powerful assembly-level debugger for Palm OS® handhelds
- A source-level debugger when using prc-tools
- A difficult tool to get started with
- Available free in Palm's "Extra Tools"

This lab will guide you through connecting to PalmDebugger's debug console and introduce -level debugging



Agenda

- Introduction to the Debugger Console
- Connecting to the Console
- Basic Console Commands
- Source Level Debugging
- Q & A



What is the Debug Console

- Debug Console is like a DOS prompt for a Palm OS® device
- Allows you to examine / manipulate databases
- You can read/write memory
- You can get information about running tasks



Where Can the Debug Console be Found

- PalmDebugger (our focus today)
- The Simulator on the Mac
- The CodeWarrior debugger



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Connecting to the Console (part 1)

1. Close or disable HotSync® Manager
2. Run the PalmDebugger application
3. Choose the proper connection from the

   got 57600 speed chosen
4. Connect your device to the proper cable/cradle



Connecting to the Console (part 2)

5. In a text field on your device, enter:
   <shortcut> . 2
   …which is a cursive, lowercase ‘el’, a period, and the number 2
6. If the console wasn’t already opened on the device, a “Ready…” message should appear in the “Console” window PalmDebugger



Connection Methods

- Things get tricky with so many different Connection methods for various OEMs
- Older Palm OS devices are easy: serial
- Handspring devices: USB is default console, but isn’t supported on Mac Debugger; hold up arrow while entering “shortcut . 2” to get serial
- Other OEMs: consult the OEMs developer documentation
- Same method as used in CW’s debugger



Connecting to the Emulator

- When you're working on the Emulator, it's like the console is always open
- The exception is if you're currently stopped in the debugger (assembly-level or source-level)



Troubleshooting Your Connection

- Making the initial connection can be tricky
- Unplugging / replugging your cable can help
- If you have a newish PalmDebugger, try entering the “ping” command in the console to verify your connection



Where Are We Now?

- Device is has the debugger console opened
- Console takes up a communication port
- The only way to close the console is by soft-resetting your device



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Entering Console Commands

- On Windows: type a command in the “Console” window and hit Enter
- On Macintosh: type a command in the “Console” window and hit Command-Return, or the Enter key on the keypad
- Think of the console window as a scratchpad—clicking on any line and hitting enter on a line executes that line



The Directory Command

- Syntax: **`dir <cardNo> [<flags>]`**
- Gives a directory of the given card, showing all databases
- Most times, there is only one card: 0
  Springboard modules introduce card 1
- Can get all sorts of attributes about your
  `dir 0 -a`”



The Export Command

- Syntax: **`export <cardNo> "<Database>"`**
- Exports the given database (resource or record database)
- Places it in PalmDebugger's "Device" folder
- Database name is case sensitive; put quotes around database name if it has a space



The Import Command

- Syntax: **`import <cardNo> "<Database.prc>"`**
- Installs the given file directly onto your device
- Source file should be in PalmDebugger's

- Can be a lot faster than using HotSync® to get a file onto your device
- Put quotes around filename if it has a space



The Delete Command

- Syntax: **`export <cardNo> "<Database>"`**
- Deletes the given database



The Heap Dump Command

- Syntax: **`hd <heapID>`**
- Shows all the memory chunks in the given heap
- On most OS 3.5 devices:
  0 – the dynamic heap
  1 – the storage heap
  2 – the ROM heap
- This command is great for finding memory leaks (as pointed out by the debug ROM)
- See Palm Knowledge Base for more info on Palm OS memory structures



The help command

- Syntax: **`help`**
- There are many more commands than we can cover here; help lists them
- Help command also gives syntax for commands



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Source Level Debugging With PalmDebugger

- Only available on Windows
- Must have used prc-tools to compile your code
- Once you have the Console open, you can load symbols using the “Source” menu
- When symbols are loaded, the “Source” window becomes useful
- You can set breakpoints when stopped in the debugger (“Break” from the “Source” menu will stop you in the debugger)



Features of the Source Level Debugger

- Standard commands: set breakpoint, step over, step into, go
- Views variables (though need to use the assembly-level window to set values)



Advantages over CodeWarrior

- Can load symbols after the fact
- Allows you to source-level debug hacks, shared libraries, springboard modules
- Can use at the same time as assembly-level debugger
- Works with prc-tools



CodeWarrior's Advantages

- Cleaner UI
- More Powerful features, such as better variable viewing
- Less Buggy
- Better integrated into development environment



For More Info

- The Console and Debugger are documented on www.PalmOS.com. Search for “console”
- There are already some tutorials about PalmDebugger’s source-level debugger on Handspring’s web site, www.Handspring.com
- A more complete tutorial on PalmDebugger will hopefully show up on Handspring’s developer web site soon



Agenda

- Introduction to PalmDebugger
- Basic Debugger Commands
- Debugging a Crash
- More Advanced Techniques
- Q & A



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